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# Sentiment analysis of news: unveiling AI's role in sustainability and no poverty (SDG1)

Robin Kunju Mol Raj<sup>1\*</sup> , Marek Vochozka<sup>1</sup> and S Sujatha<sup>2</sup>

\*Correspondence:

Robin Kunju Mol Raj  
32110@mail.vstecb.cz  
<sup>1</sup>Institute of Technology and  
Business in České Budějovice,  
School of Expertness and Valuation,  
Okružní 517/10,  
České Budějovice  
37001, Czech Republic  
<sup>2</sup>Hindustan College of Arts &  
Science, Coimbatore 641028, Tamil  
Nadu, India

## Abstract

Media play an essential role in shaping public attitudes, behaviors, and long-term societal developments. Although AI is often addressed in the context of technical advancement, its role in media discourse on Sustainable Development Goal 1 (SDG1 - No Poverty) is underexplored. Media sentiment towards AI and poverty is researched here through sentiment analysis, topic modeling, and predictive modeling of global news headlines on the GDELT database. This study addresses this gap by examining worldwide media sentiment and narrative trends on AI and poverty from 2017 to 2025. Sentiment classification used VADER, TextBlob, and BERT models, with a net positive sentiment reported in AI-poverty stories. Latent Dirichlet Allocation (LDA) topic modeling revealed labor workforce automation and sustainable economic transformation to be the most discussed themes. In addition, Long Short-Term Memory (LSTM) neural networks were employed to predict sentiment trends. The model demonstrated low predictive validity ( $R^2 = 0.0124$ ; Pearson correlation = 0.1223), suggesting that additional contextual variables are needed. Notably, BERT performed a more accurate detection of subtle changes in sentiment than VADER and TextBlob, which tended to make more coarse categorizations. The LDA model further revealed consistent discussions surrounding government policy, labor challenges, and ethical concerns about automation. These findings highlight the polarized nature of AI-related media coverage in poverty contexts and the value of sentiment analysis in assessing AI's perceived socio-economic role. Our study contributes to the literature by combining computational approaches to evaluate sustainability-oriented media narratives.

**Keywords** Artificial intelligence (AI), Poverty, Sentiment, LDA, News, Automation, BERT, VADER, TextBlob

## 1 Introduction

Artificial Intelligence (AI) is increasingly recognized as a transformative force in addressing global socio-economic challenges, particularly in poverty alleviation [51]. AI applications have expanded to include prediction, analysis, and targeted intervention strategies aimed at reducing poverty levels. As part of international development efforts, AI is being integrated into policy frameworks to help achieve Sustainable Development Goal 1 (SDG1), which calls for the eradication of poverty in all its forms [36]. By leveraging



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machine learning algorithms, data analytics, and automation tools, AI supports better decision-making, optimizes resource allocation, and contributes to the socio-economic development of vulnerable populations.

The role of media is pivotal in shaping public perception and influencing policy dialogues around AI and poverty. News coverage not only informs citizens but also plays a part in legitimizing specific narratives and technologies. Sentiment analysis of media content provides a methodical approach to assess how AI is framed about economic development, employment, and social justice [42]. Examining media sentiment allows us to better understand both the opportunities and anxieties that AI introduces in the context of poverty reduction.

AI has demonstrated significant potential in predicting poverty through a variety of methods. A review of literature since 2016 shows progress in AI-driven poverty estimation. Algorithms such as Random Forest have achieved high accuracy in poverty prediction tasks [49], and the adaptability of AI models to various data environments has made them increasingly reliable for socio-economic analysis [76]. However, the link between AI and poverty reduction is multifaceted and deeply context-dependent [57]. Sentiment analysis of news and reports related to AI's role in poverty alleviation enriches this understanding by highlighting public expectations and potential risks [47].

Emerging technologies such as AI, big data, and the Internet of Things (IoT) provide new pathways for data-driven policy design [1]. For example, combining AI with satellite imagery improves poverty estimation but introduces concerns about interpretability and transparency [23]. While AI may enhance public service delivery, it can also perpetuate biases and marginalization when implemented through deficit-based models [72]. The success of AI interventions is also shaped by financial flows, such as remittances [50], and methodological factors [22]. Though challenges remain, AI continues to show promise in transforming public services in low-resource settings when deployed ethically [78].

To address data limitations in conventional poverty measurement, researchers have explored alternative sources, including e-commerce data and health records, processed through machine learning models like XGBoost and 1D CNNs [3, 44, 80]. However, these innovations raise new ethical concerns, such as poverty stigma [43], algorithmic bias due to health data poverty [62], and disparities in access to clean energy [55]. Evidence from South Africa, for example, suggests that AI investments supported by educational policy can raise household consumption and reduce poverty [4].

Understanding how the media portrays AI in the context of poverty is vital for government agencies and development actors [83]. Media sentiment analysis allows policymakers to assess public engagement with AI, understand regional variations, and evaluate the alignment between technological developments and SDG1 objectives. Tools such as VADER, TextBlob, and BERT enable researchers to quantify sentiment from diverse sources [45]. By applying these techniques to cross-platform news articles, researchers can uncover narrative patterns, evaluate positive, negative, and neutral framing, and identify knowledge gaps in public discourse [77].

While prior studies have addressed AI's potential for achieving various SDGs, few have critically examined how AI is portrayed in the media in relation to SDG1 (No Poverty). This research addresses that gap by analyzing global news sentiment toward AI and poverty from 2017 to 2025. The novelty of this work lies in its integration of AI discourse analysis with sustainability metrics, offering a new perspective for understanding how

public narratives align or misalign with development goals. This study offers valuable insights for policymakers, media professionals, and development stakeholders interested in how AI's image is constructed within the global poverty dialogue.

This paper contributes to the literature by connecting three key domains: AI and sustainability [8, 59, 64, 65, 79], media influence on technology perceptions [54, 81], and sentiment analysis as a methodological lens to study policy-relevant narratives [10, 66, 71]. Media framing of AI as both a potential solution and a source of risk informs public expectations and policy responses [19]. This leads us to the first research question:

**RQ1** How do media narratives frame AI's role in poverty, and what are the dominant sentiment trends over time?

Discussions surrounding AI and poverty often alternate between its potential for economic inclusion and fears of automation-driven job loss [37]. Analyzing these themes and how they are interpreted by various sentiment analysis models [12, 13, 31, 38, 74] offers further insight into public opinion [15] and narrative construction [16, 25, 33, 46, 75]. This motivates our second research question:

**RQ2** What are the key themes in AI and poverty-related media discussions, and how do different sentiment analysis models interpret them?

## 2 Review of literature

### 2.1 AI in poverty prediction and measurement

Artificial Intelligence (AI) technology increasingly allows poverty estimation through the use of machine learning models, satellite data, and big survey data. Usmanova et al. [76] reviewed 22 studies and found Random Forest as one of the most frequent models for poverty prediction since 2016. These models yield greater accuracy and effectiveness when paired with broad-spectrum datasets. Khare and Jadhav [34] also proposed a hybrid approach through the combination of Sentinel-2 satellite imagery and household survey data, reducing the errors of poverty estimation from 4.09 to 3.71% through a Proxy Means Test. Hoogstra et al. [29] also built on the method by combining demographic data with spatial satellite imagery through a Vector Autoregression (VAR) ensemble model, subjecting consideration for spatial interconnectedness and neighborhood patterns of poverty. Despite the great strides that such studies exhibit, they inevitably raise questions of how to handle limitations such as algorithmic bias, model generalizability, and equitable access to AI tools [21, 51]. While these contributions advance technical knowledge, they frequently fail to explore how such models work in real-world policy situations or how they accommodate vulnerable, underrepresented populations.

### 2.2 Media framing and public perception of AI

Social perception regarding AI and how it can be used to end poverty is greatly influenced by the way it is framed by the media. Garvey and Maskal [19], and Zuiderwijk et al. [83] demonstrate that whether AI is cast as an opportunity or a threat can influence policy support and public trust. Mao et al. [47] analyze AI economic sentiment in trade policy uncertainty and financial technology with emphasis on the role of external controls on sentiment. Krugmann and Hartmann [39] analyze development in sentiment

analysis under generative AI with a note on its effect on social and economic welfare assessment frameworks. Research by Nazneen et al. [60], Scheyvens and Hughes [70], and Zhang et al. [82] indicates the lack of global coherence in how AI is presented in SDG1 contexts. These findings underscore the need for a more consistent and nuanced understanding of AI's societal impact. Furthermore, few studies examine how media-driven public perception leads to real policy changes or resource allocations, leaving a vacuum between debate and implementation.

### 2.3 Sentiment analysis methods in socioeconomic research

Sentiment analysis, being one of the core aspects of NLP, is a vital resource for examining the discourse of AI in the context of socio-economic concerns. Transformer models and deep learning frameworks have outperformed the previous methods in detecting subtle sentiment in challenging and multilingual corpora [35, 52]. BERT [17, 69], VADER [5, 6, 26, 48], and TextBlob [2, 9] are widely used models for sentiment categorization of social media and news content. While such tools offer reliable emotion detection, Hartmann et al. [24] recommend that very specific models, like transfer learning methods, may have interpretability problems, especially in policy-related subjects. Integration of sentiment analysis with neural network models such as LSTM [40] allows prediction of sentiment patterns over time, linking perception analysis with predictive modeling.

### 2.4 Limitations, bias, and gaps in existing literature

Despite the growing literature, there are several gaps. Curto et al. [11] think that social scientists must be involved to integrate fairness and ethical frameworks in deployed AI systems. Models such as Word2vec have been shown to discriminate against lower classes, indicating a system issue in the data employed to train them and the assumptions of the model. Mussida and Sciulli [56], and Nahar [57] suggest that while AI is destined to influence productivity and inclusion, studies remain short of a resilient, cross-country temporal projection of its influence on SDGs. Moreover, studies remain biased towards technical performance rather than socio-political ramifications, overlooking the broader systemic impact that AI may perpetuate or counteract. Furthermore, the existing work seldom bridges the methodological gap between sentiment extraction and long-term SDG projection, resulting in fragmented evidence bases.

### 2.5 Bridging the gaps: toward an integrated framework

This study builds on these findings by combining sentiment analysis and SDG-focused AI discourse analysis for the evaluation of public discourse in the poverty alleviation landscape. Drawing on classification using BERT, VADER, and TextBlob, and trend forecasting with LSTM, the approach responds to scholarly calls for more consolidated and interpretable tools [15, 62, 66]. The model enables both descriptive and predictive insights, improving our understanding of how AI is presented in SDG1-related news and offering an extensible template for cross-domain policy analysis.

Table 1 below summarizes key studies related to AI, sentiment analysis, and poverty reduction, highlighting their methodologies, focus areas, and primary findings.

**Table 1** Overview of key studies on AI, sentiment analysis, and poverty reduction

| Author (s) and Year | Methodology                                                                  | Focus Area                                                    | Key Findings                                                                                                                      |
|---------------------|------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| [32]                | Theoretical analysis using moral sentiment theory                            | Ethics of global poverty and emotion's role in moral judgment | Emotions enhance ethical reasoning and motivate action in response to global injustice                                            |
| [27]                | Theoretical critique and reformulation                                       | Motivation gap in addressing global poverty                   | Proposes a dialogical model of sentimental education to inspire political-not just charitable-action                              |
| [61]                | Survey of 440 Singaporeans                                                   | Public perceptions and attitudes toward poverty               | Sympathy linked to narrow definitions; views vary by political and professional affiliation                                       |
| [14]                | Mixed-method Twitter analysis using NodeXL Pro                               | Food poverty discourse during COVID-19                        | Tweets linked rising food poverty to unemployment, panic buying, and system disruptions; called for long-term policy responses    |
| [63]                | Bibliometric analysis of 253 studies                                         | AI's role in poverty alleviation (SDG 1)                      | Identified key trends, mechanisms, and gaps; highlights AI's potential in addressing poverty                                      |
| [24]                | Meta-analysis of 272 datasets comparing lexicon and transfer learning models | Sentiment analysis model performance and interpretability     | Transfer learning models are more accurate but less interpretable than lexicon-based methods; context matters.                    |
| [18]                | Topic modeling & sentiment analysis on 6.5 M tweets                          | COVID-19 discourse in U.S. & Brazil (English & Portuguese)    | Identified top 10 topics; sentiment trends reflected media and public reactions over time                                         |
| [58]                | Sentiment classification on 90k COVID-19 tweets (COVIDSENTI dataset)         | Public opinion on COVID-19 and misinformation                 | Negative sentiment influenced public behavior; sentiment shifted over time; highlights need for policy on social media moderation |
| [28]                | Emotion-based sentiment analysis using Syuzhet and NRC lexicons              | Emotional content in 2016 U.S. presidential campaign speeches | Fine-grained emotion analysis reveals richer insights than binary sentiment polarity                                              |
| [80]                | ML models using e-commerce data with feature selection                       | Poverty prediction in Indonesia                               | SVR with F-score selection outperformed others; shows potential of e-commerce data as poverty proxy                               |
| [1]                 | Conceptual review                                                            | Industry 4.0 & poverty in Africa                              | Tech offers potential, but needs strong policy and capacity support                                                               |
| [21]                | Case study of Ecuador's Social Registry                                      | AI and datafication in digital welfare                        | Highlights risks of exclusion and inequality in AI-driven poverty management                                                      |
| [30]                | Granger causality on Twitter and news sentiment data.                        | Role of news sentiment in shaping social media sentiment.     | Mutual influence; CNN swayed by Clinton supporters, Fox News influenced Trump supporters                                          |

This table positions the current study within the broader research landscape and highlights how it uniquely combines sentiment analysis of AI-related news with an SDG1 (No Poverty) focus

3 Data and methodology

Sentiment analysis, topic modeling, and predictive modeling are used in this study's quantitative and computational investigation of media narratives about AI and poverty. Examining sentiment patterns, identifying important conversation topics, and predicting sentiment changes over time are the goals. Large-scale news items are processed, and insights are extracted using a Natural Language Processing (NLP) based methodology. Latent Dirichlet Allocation (LDA) is utilized for topic modeling to identify dominating topics, whereas sentiment analysis models (VADER, TextBlob, and BERT) are employed to evaluate media perception. Furthermore, to forecast future sentiment patterns, a Long Short-Term Memory (LSTM) neural network is used; performance measures are used to assess the network's accuracy. The study employs a data-driven approach, interpreting results via the use of visual analytics and machine learning approaches.

This multi-faceted approach ensures a comprehensive understanding of AI’s portrayal in media discourse over time. This study did not involve human participants or the use of human tissue samples. All data were collected from publicly available news sources. Consequently, ethical approval and informed consent were not applicable (Table 2).

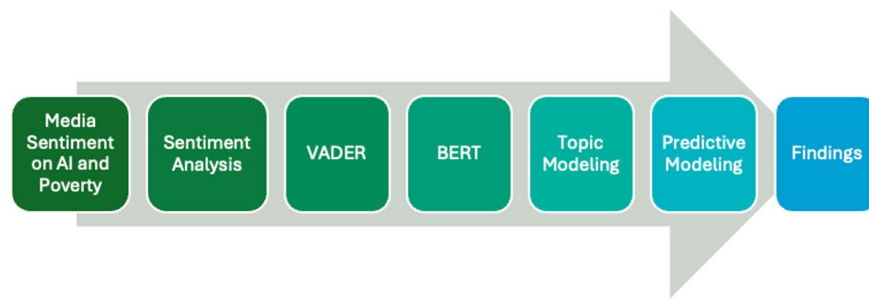
The Global Database of Events, Language, and Tone (GDELT) Project ([www.gdeltproject.org](http://www.gdeltproject.org)), a publicly accessible database that tracks news media coverage worldwide, provided the dataset for this study. A longitudinal examination of media narratives about AI and poverty is made possible by the data, which covers the period from January 1, 2017, to February 8, 2025. No projections or future sentiment simulations were used; only actual, real-time data up to February 8, 2025, were included in the analysis. After preprocessing more than 80,000 news articles from various digital platforms, a total of 11,492 relevant articles were retained for analysis. An initial corpus of news stories gathered from multiple digital platforms was filtered based on language (English only), relevance to AI and poverty-related keywords, and the elimination of duplicates or incomplete data. This modification yielded a final dataset of 11,492 articles appropriate for sentiment analysis. Non-English articles were eliminated to maintain language integrity and prevent sentiment distortion caused by translation. GDELT was chosen because of its extensive, real-time, worldwide media monitoring [41], which includes multilingual coverage of blogs, news stories, and other digital sources. For the data preprocessing, to ensure consistency and reliability, the following preprocessing steps were applied. At first, for the data extraction, the articles related to AI and poverty were filtered using relevant keywords and event codes from GDELT’s database. From the data cleaning, the duplicate entries and irrelevant articles were removed. Thereafter, text Processing, which contains Tokenization, stopword removal, and lemmatization, was performed to standardize text for NLP analysis. After that, the sentiment labeling, each article was assigned sentiment scores (positive, negative, or neutral) using VADER, TextBlob, and BERT sentiment analysis models. The cleaned and structured dataset was then used for sentiment analysis, topic modeling, and predictive modeling, ensuring accurate and meaningful insights into AI’s role in poverty-related discourse. Figure 1 shows the methodological framework for analyzing media sentiment on AI and Poverty.

All sentiment analysis was implemented in Python 3.10 using the following libraries: for BERT, we used the bert-base-uncased model via the HuggingFace Transformers library (v4.35.2). Sentiment classification was performed using the pre-trained model `nlptown/bert-base-multilingual-uncased-sentiment`, which provides five-level sentiment outputs (1 to 5 stars). No fine-tuning was applied. For the VADER, implemented using the `nlk` package (v3.8.1) with the default compound score threshold (compound > 0.05 = positive, < -0.05 = negative, otherwise neutral). And for the TextBlob,

**Table 2** Data collection and processing steps

| Step                | Description                                                                    |
|---------------------|--------------------------------------------------------------------------------|
| Data source         | Global Database of Events, Language, and Tone (GDELT), 2017–2025 coverage      |
| Data extraction     | Filtered news articles using AI- and poverty-related keywords and event codes  |
| Data cleaning       | Removed duplicates and non-relevant articles                                   |
| Text preprocessing  | Applied tokenization, stopword removal, and lemmatization                      |
| Sentiment labeling  | Assigned sentiment scores using VADER, TextBlob, and BERT                      |
| Analysis techniques | Used the processed data for sentiment analysis, topic modeling, and prediction |

Note: This table summarizes the sequential steps taken to prepare and analyze the news dataset retrieved from the GDELT Project. Each step was designed to ensure data consistency, relevance, and readiness for natural language processing and machine learning tasks



**Fig. 1** Methodological Framework for Analyzing Media Sentiment on AI and Poverty Note: This flowchart outlines the step-by-step approach used in the study, beginning with sentiment analysis using VADER and BERT, followed by topic modeling with LDA, predictive modeling using LSTM, and ending with result interpretation.

Applied via the textblob library (v0.17.1) using the default sentiment polarity scoring. A polarity above 0.1 was classified as positive, below  $-0.1$  as negative, and in between as neutral. All computations were performed on a machine with an Intel Core i7 processor, 16 GB RAM, and Ubuntu 22.04 LTS. Results were validated across multiple runs to ensure consistency in sentiment classification outcomes.

**VADER sentiment analysis:** VADER (Valence Aware dictionary and sEntiment Reasoner) assigns sentiment scores based on a lexicon of words that have predefined sentiment intensities. The compound sentiment score (CS) is calculated using the following formula.

$$CS = \frac{\sum (v \times w)}{\sqrt{\sum (v \times w)^2}} \quad (1)$$

Where the  $v$  (valence) represents the sentiment intensity of each word in the text (from the VADER lexicon) and the  $w$  (weight) the word's importance, adjusted by punctuation, capitalization, negation, and other heuristics.

**TextBlob Sentiment Analysis:** TextBlob calculates two sentiment scores for a given text: they are Polarity and Subjectivity. The polarity score ( $P$ ) is computed as the weighted average of sentiment scores of individual words in the text:

$$P = \frac{\sum_{i=1}^n \text{score}(w_i) \times \text{weight}(w_i)}{\sum_{i=1}^n \text{weight}(w_i)} \quad (2)$$

Where  $\text{score}(w_i)$  is the sentiment polarity of word  $w_i$  (ranging from  $-1$  (negative) to  $+1$  (positive)).  $\text{weight}(w_i)$  is the word importance based on word position, negations, and modifiers.  $n$  is the total number of words in the text.

The subjectivity score ( $S$ ) calculated as follows:

$$S = \frac{\sum_{i=1}^n \text{subjectivity}(w_i)}{n} \quad (3)$$

Where  $\text{subjectivity}(w_i)$ , is the subjectivity score of word  $w_i$  (ranging from 0 (objective) to 1 (highly opinionated)) and  $n$  is the total number of words in the text. TextBlob assigns equal weight to all words when computing subjectivity, meaning that every word contributes equally to the overall subjectivity score.

**BERT Sentiment Analysis:** BERT (Bidirectional Encoder Representations from Transformers) is a deep learning-based NLP model that processes text using self-attention and contextual embeddings. Unlike VADER and TextBlob, which use predefined lexicons, BERT learns sentiment contextually from large datasets. Each input sentence is tokenized and converted into a numerical embedding matrix:  $X = \{x_1, x_2, \dots, x_n\}$ , where  $X$  is the input sequence,  $x_i$  represents each tokenized word in the text. The token embeddings, segment embeddings, and positional embeddings are summed to create a final input vector. Each token's contextual representation is computed using multi-head self-attention:

$$Z = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V. \quad (4)$$

Where  $Q = XW_Q$ ,  $K = XW_K$ ,  $V = XW_V$  are the query, key, and value matrices, obtained by multiplying the input embedding  $X$  with learnable weight matrices  $W_Q$ ,  $W_K$ ,  $W_V$ .  $d_k$  is the dimensionality of the key vectors (used for scaling). The softmax function normalizes attention scores. The final representation of the [CLS] token (first token in BERT input) is passed through a fully connected layer with a softmax activation to predict sentiment categories:

$$Z = \text{softmax} (W_h \cdot h_{CLS} + b). \quad (5)$$

Where  $h_{CLS}$  is the BERT embedding of the [CLS] token (which captures the entire sentence's meaning). And  $W_h$  and  $b$  are trainable weights and biases. The softmax function converts logits into probabilities for sentiment classes (positive, negative, neutral):

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}. \quad (6)$$

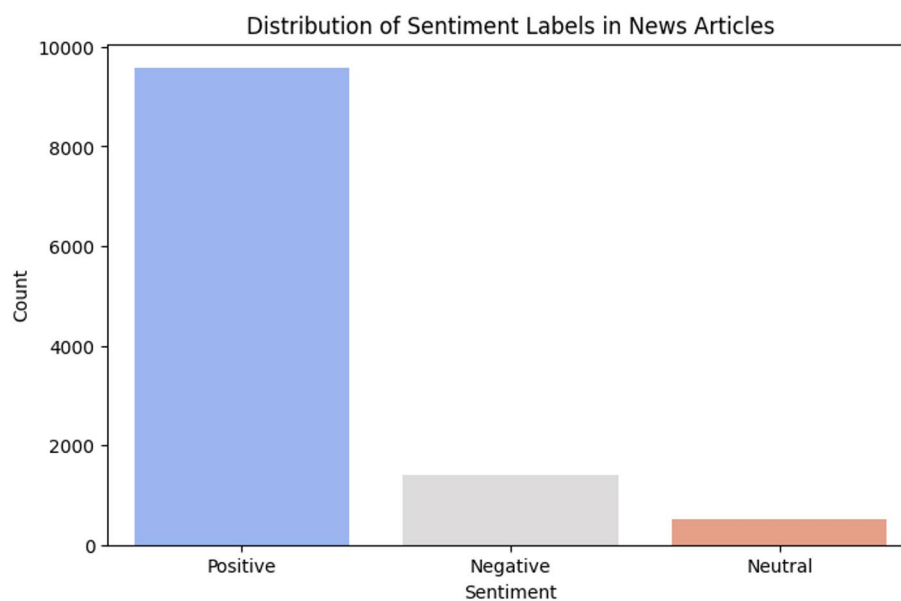
BERT's context-aware embeddings allow it to capture nuanced sentiment better than lexicon-based models like VADER and TextBlob.

The data consists of 11,486 articles published between January 1, 2017, and February 8, 2025, offering a rich perspective of the media discussions around the words AI and poverty. The sentiment in AI and poverty is imposed positively, based on the mean sentiment and polarity score (0.105) and mean VADER score (0.685) indicators. Although there is a conventionally positive vein throughout these articles, not only positivity, but also there is negativity (min:  $-0.700$ , VADER min:  $-0.999$ ) in that the resounds of the discussion of AI, restrictively as it bonds to labor markets and inequality, are closing that confabulation. This moderate subjectivity score (mean: 0.398, max: 1.000) indicates that while news articles report on facts, they also tend to have opinion-based content, which can be useful to influence public opinion through both data around the event and the personal interpretation of it. Although there is some variability around the intensity of the sentiment (VADER std: 0.589), the data shows that AI is predominantly framed positively in articles, and a subset for which, on average, seems to question whether there are risks and ethics that need to be taken into consideration. An study of the sentiment distribution found a clear skew toward positive sentiment in the data. This tilt persists over various years and techniques (VADER, TextBlob, BERT), possibly reflecting innate media optimism or a reporting bias in publicly available news headlines (Table 3).

**Table 3** Descriptive statistics of sentiment, polarity, subjectivity, and VADER score

| Statistic | Sentiment | Polarity | Subjectivity | VADER Score |
|-----------|-----------|----------|--------------|-------------|
| Count     | 11,486    | 11,486   | 11,486       | 11,486      |
| Mean      | 0.105     | 0.105    | 0.398        | 0.685       |
| Min       | − 0.700   | − 0.700  | 0.000        | − 0.999     |
| 25%       | 0.063     | 0.063    | 0.357        | 0.788       |
| 50%       | 0.103     | 0.103    | 0.412        | 0.987       |
| 75%       | 0.144     | 0.144    | 0.458        | 0.997       |
| Max       | 1.000     | 1.000    | 1.000        | 1.000       |
| std       | 0.085     | 0.085    | 0.112        | 0.589       |

Note: The statistical summary of sentiment and topic analysis signifies that AI and poverty-related news articles generally have a positive sentiment, with a mix of factual and subjective reporting, while topic clustering indicates a strong focus on economic transformation and workforce implications



**Fig. 2** Distribution of Sentiment Labels in News Articles Note: The graph demonstrates a highly polarized media sentiment on AI and poverty, with overwhelmingly positive coverage (9577 articles), notably negative concerns (1397 articles), and minimal neutral representation (512 articles), emphasizing the need for a more balanced discourse.

**Topic Modeling (LDA):** To uncover the key themes in AI and poverty-related discussions, this study employs Latent Dirichlet Allocation (LDA), a probabilistic topic modeling technique. LDA identifies hidden topics in a collection of news articles by analyzing word distributions and assigning topics to documents based on the likelihood of word co-occurrence. LDA assumes that each document ( $d$ ) is a mixture of topics ( $z$ ), and each topic ( $z$ ) is a distribution of words ( $w$ ). The probability of a word belonging to a topic follows a Dirichlet distribution. The LDA model can be mathematically expressed as:

$$P(w \mid \alpha, \beta) = \int_{\theta} \sum_z P(w \mid z, \beta) P(z \mid \theta) P(\theta \mid \alpha) d\theta \quad (7)$$

Where  $w$  is the observed words in the dataset (news articles),  $z$  is the hidden topic assignments for words,  $\theta$  is the per-document topic distribution and  $\alpha, \beta$  are the Dirichlet priors for topic and word distributions. LDA uses Gibbs Sampling to



**Fig. 3** Sentiment Trend Over Time Note: The graph demonstrates the sentiment trend over time for news articles on AI and Poverty from 2017 to 2025, showing a highly polarized distribution with a concentration of positive sentiment scores near one and periodic negative sentiment spikes near  $-1$ , indicating fluctuating media narratives and increasing discourse on AI's impact on poverty.

estimate these parameters and infer the most likely topic distributions for each article. The implementation process, containing preprocessing, Model training, and the topic interpretation.

**Predictive modeling (LSTM):** A Long Short-Term Memory (LSTM) neural network was used to predict sentiment patterns in the future. A recurrent neural network (RNN) architecture called LSTM was created especially to process sequential data, which makes it appropriate for time-series sentiment analysis. Sentiment scores from VADER, TextBlob, and BERT were utilized as input features in the feature selection step of the data preparation procedure. With prior sentiment ratings acting as inputs for future forecasts, the dataset was organized chronologically in a time-series style. The dataset was divided between 80% training and 20% testing data, and sentiment ratings were standardized between 0 and 1 to improve model performance. The design of the LSTM model comprised a fully connected dense output layer to produce sentiment score predictions, dropout layers to avoid overfitting, and LSTM layers to capture sequential relationships. The Mean Squared Error (MSE) loss function was used to train the model across several epochs, and the Adam optimizer was used to optimize it. Prediction accuracy was measured using Mean Absolute Error (MAE), while the size of prediction errors was evaluated using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). Additionally, the link between the predicted and real sentiment scores was measured using Pearson correlation, and the  $R^2$  score was computed to assess how well sentiment patterns were described.

#### 4 Results

The sentiment analysis of news articles on AI and Poverty discloses an extremely polarized media landscape, with an overwhelmingly positive sentiment (9577 articles), an eminent negative sentiment (1397 articles), and minimal neutral coverage (512 articles). The takeover of positive sentiment indicates that AI is widely perceived as a transformative tool for poverty alleviation, enhancing access to education, healthcare, and financial inclusion. At the same time, the presence of negative sentiment emphasizes the concerns about job displacement, ethical risks, and AI biases distinctively affecting low-income

communities. The finite number of neutral articles denotes that media narratives on AI's role in poverty are often vigorously opinionated rather than stable, certainly influencing public perception and policy decisions. Figure 2 and 3below addresses the distribution of Sentiment Labels in the News articles..

The sentiment trend over time graph demonstrates the variation in sentiment scores of news articles related to AI and Poverty from 2017 to 2025. The vertical spread of sentiment values between  $-1$  (negative) and  $1$  (positive) signifies a high degree of inconsistency, indicating that media documentation has been highly polarized throughout the years. The dense clustering of positive sentiment scores near  $1$  indicates that news articles overwhelmingly favor AI's role in poverty alleviation, while the negative sentiment spikes near  $-1$  emphasize periodic concerns, potentially linked to events such as job displacement fears, ethical issues, or technological failures. The rising density of data points over time signifies a growing volume of discussions on this topic, replicating AI's expanding influence on economic and social issues.

The word clouds for positive and negative sentiment articles or news on AI and Poverty emphasize contrasting media narratives. In the positive sentiment word cloud, key terms such as “need,” “work,” “automation,” “technology,” and “future” indicate a promising outlook on AI’s role in enhancing economic opportunities, job creation, and service efficiency. The presence of words like “government” and “provides” signifies that these articles emphasize policy support and AI-driven advancements as solutions to poverty. In contrast, the negative sentiment word cloud is conquered by words like “automation,” “job,” “poverty,” “people,” and “will,” replicating concerns about job displacement, economic inequality, and AI’s potential to widen the digital divide. The prominence of “robots” and “percent” indicates fears related to wage loss, workforce automation, and economic instability. These insights signify a polarized discourse, where AI is either viewed as a transformational force for poverty alleviation or a disruptive factor threatening livelihoods (Fig. 4).

The bar chart demonstrates the sentiment distribution of news articles from the top 20 news sources covering AI and poverty content. The plurality of articles displays a positive sentiment (red bars), with PRNewswire and OpenPR contributing the supreme number of positively framed articles. This indicates that press release platforms and business-focused outlets are promoting AI's role in economic and social development. However, some sources, such as Digital Journal and Business Wire, display a relatively higher proportion of neutral (gray) and negative (blue) sentiment, indicating a more balanced or critical discussion of AI's implications for poverty. The presence of negative sentiment across multiple sources indicates that while AI is generally noticed positively,



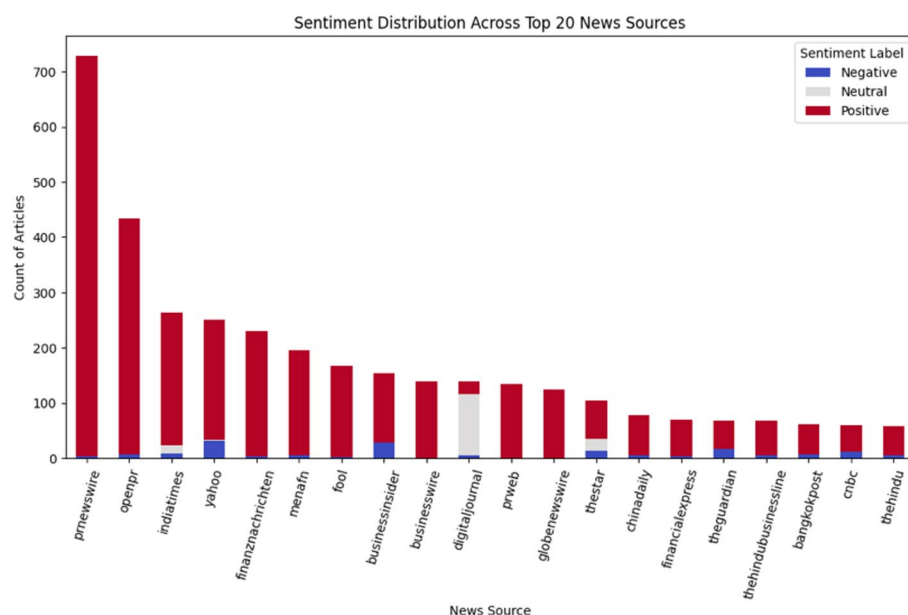
**Fig. 4** Word Cloud for Positive and Negative Sentiment Articles Note: The word clouds visualize the key terms associated with positive and negative sentiment articles on AI and Poverty, revealing a strong contrast between optimism about AI-driven economic growth and concerns about job displacement and inequality.

there are continuing interests regarding automation, job displacement, and economic disparities (Figs. 5 and 6).

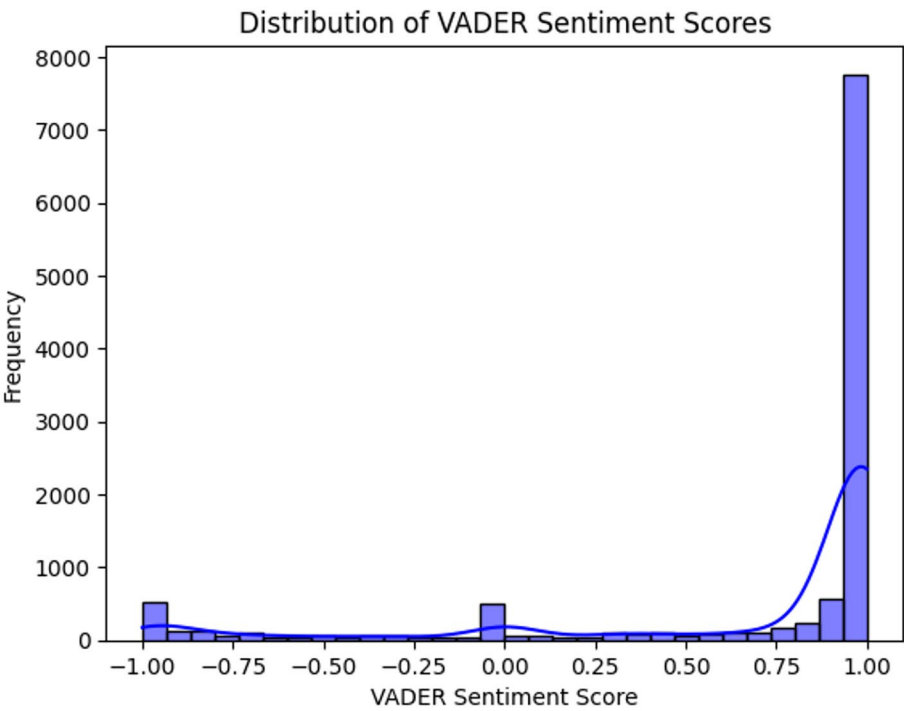
The histogram demonstrates the distribution of VADER sentiment scores in news articles related to AI and Poverty. The plurality of sentiment scores is vigorously skewed towards +1, indicating a supremacy of highly positive sentiment in the dataset. There are significantly fewer articles with neutral (0) or negative (-1) sentiment, indicating that media narratives predominantly depict AI's role in poverty alleviation in a promising manner. The small peaks near -1 and 0 replicate instances where AI is associated with negative socio-economic consequences, such as job losses or digital inequalities. The KDE (Kernel Density Estimate) curve reinforces this distribution, displaying a strong positive sentiment bias with limited variance in neutrality and negativity.

The scatter plot displays the relationship between polarity (sentiment score) and subjectivity in news articles related to AI and Poverty. The plurality of articles (blue points) displays positive polarity (right of 0), confirming the previous findings that media coverage is highly supportive of AI's role in poverty alleviation. Besides, most articles have moderate subjectivity values, indicating a mix of factual reporting and opinion-based writing. The negative sentiment articles (red points) are more propagated across the lower polarity range, but they remain fewer in number. Neutral articles (gray points) cluster near zero polarity and low subjectivity, portraying objective news content. The generic trend indicates that positive articles are more subjective in the sense that they often include opinions or persuasive narratives. In contrast, negative and neutral articles tend to be more fact-driven and balanced (Figs. 7 and 8).

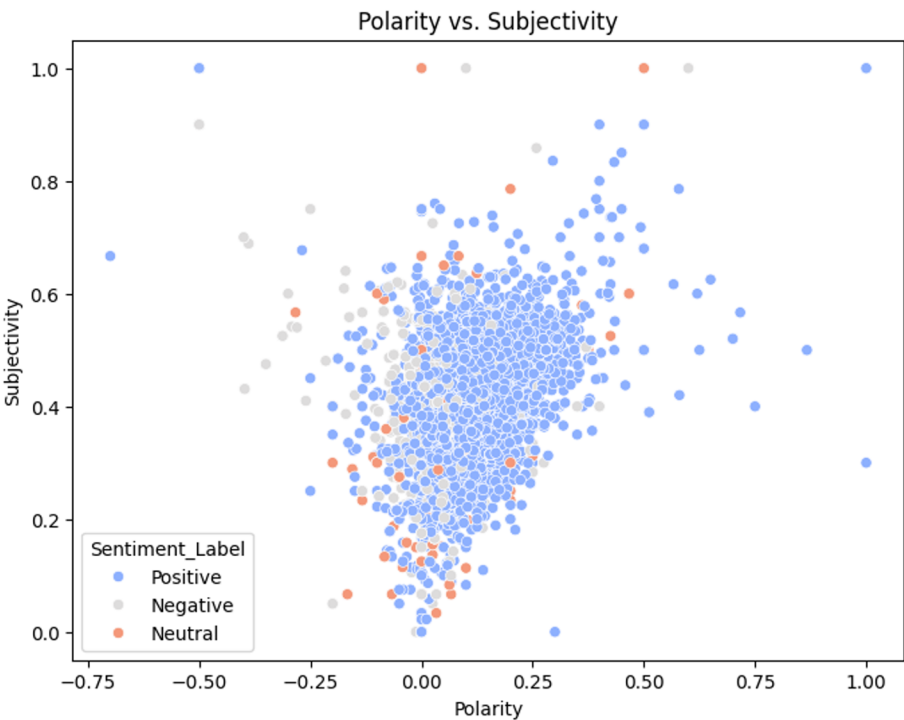
For news items on AI and poverty, the scatter plot contrasts sentiment ratings given by two popular sentiment analysis tools, TextBlob and VADER. Both models concur on a categorization of primarily positive sentiment, as seen by the substantial clustering of positive sentiment (blue spots) in the upper half. The neutral (orange) and negative



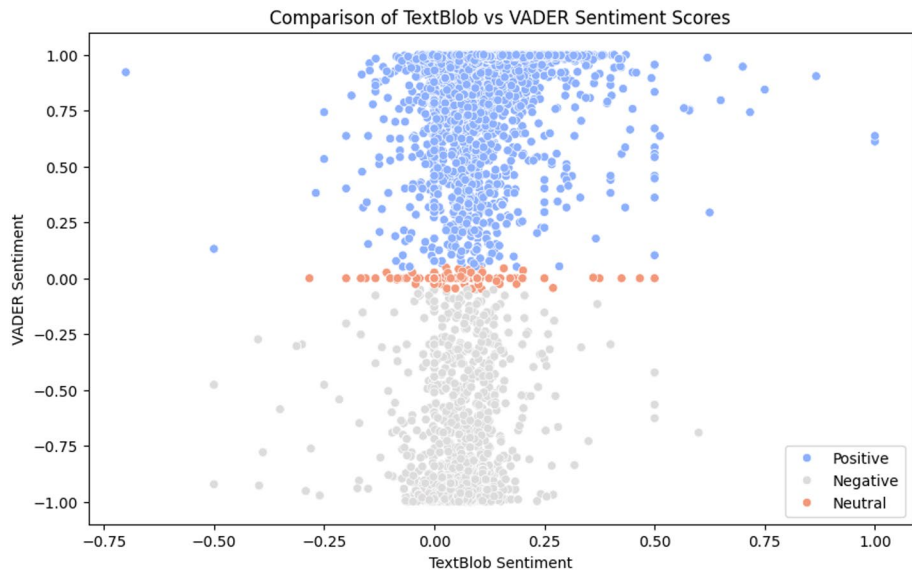
**Fig. 5** . Sentiment Distribution Across Top 20 News Sources Note: The chart displays the distribution of sentiment-labeled news articles across the top 20 news sources, showing strong supremacy of positive sentiment, particularly in press release platforms like PRNewswire and OpenPR, while a few sources display a more neutral or negative stance on AI's role in poverty.



**Fig. 6** Distribution of VADER Sentiment Scores Note: The histogram shows the distribution of VADER sentiment scores, emphasizing a strong bias toward positive sentiment, with fewer instances of neutral and negative sentiment, indicating a predominantly promising media portrayal of AI’s impact on poverty.



**Fig. 7** Polarity v/s Subjectivity in the case of Sentiment Label Note: The scatter plot demonstrates the relationship between sentiment polarity and subjectivity in AI & Poverty news articles, showing a strong concentration of positive sentiment with moderate subjectivity, while negative and neutral articles appear less frequently and are more objectively framed.



**Fig. 8** Comparison of TextBlob v/s VADER Sentiment Scores Note: The scatter plot displays the comparison between TextBlob and VADER sentiment scores, showing that both models generally align on positive sentiment but differ in classifying neutral and negative sentiment, with VADER capturing more extreme polarity than TextBlob.

**Table 4** Correlation between TextBlob and VADER sentiment scores

|                    | Textblob sentiment | Vader sentiment |
|--------------------|--------------------|-----------------|
| Textblob sentiment | 1.0                | 0.292           |
| Vader sentiment    | 0.292              | 1.0             |

Note: The table presents the correlation between TextBlob and VADER sentiment scores, showing a moderate positive correlation (0.29), indicating that while both models align partially, they differ in their interpretation of sentiment polarity and intensity.

(gray) sentiment ratings, however, clearly differ from one another. TextBlob tends to give more neutral values, whereas VADER offers a wider range of scores, including stronger polarity. The fact that the negative sentiment cluster (gray) is more widely distributed suggests that TextBlob and VADER have distinct interpretations of negative sentiment, with VADER capturing polarity with more intensity. This comparison highlights how sentiment classification techniques differ, with VADER being more context and emotion-sensitive and TextBlob leaning towards a neutral or softer classification.

TextBlob and VADER sentiment scores have a moderately positive correlation (0.29) according to the correlation matrix, which means that although there is some alignment between the two sentiment analysis models, there are also notable categorization disparities. While TextBlob tends to award more neutral ratings, VADER is more context-sensitive and captures higher sentiment polarity, as seen by the comparatively low correlation between the two sentiment analysis tools. This highlights how crucial it is to use a variety of sentiment models for robustness because depending only on one model may result in bias in sentiment categorization (Table 4).

The mean topic (1.693) and cluster (2.193) values, which show that workforce automation and economic transformation are the most talked-about topics, replicate the topic and sentiment analysis statistics, which show that discussions about AI and poverty are largely focused on a few dominant themes. An organized distribution of tales on AI is further supported by the BERT subject classification (mean: 1.174, maximum: 3.000). The considerable positive bias in media coverage is further supported by the high

average BERT score (0.943) and sentiment score (0.712), which show that most people see AI’s contribution to poverty favorably. Nonetheless, a certain amount of polarization is indicated by the existence of negative sentiment (min sentiment score: – 1.000) and variance in subject clustering (std: 1.266), especially in conversations about labor and policy. Despite high sentiment scores, generic, the variability in sentiment intensity (std: 0.669) indicates that while AI is mostly seen as transformative, some skepticism remains regarding its broader economic and societal impacts (Table 5).

For news stories on AI and poverty, the bar graph contrasts TextBlob and VADER sentiment labels with BERT-based sentiment classification. The pattern of encouraging media attention is confirmed by the fact that the majority of stories categorized as “Positive” by BERT are also classed as such by TextBlob and VADER. However, a sizable percentage of the “Negative” articles identified by BERT are classified as Neutral or Positive by TextBlob and VADER, demonstrating variations in emotion recognition techniques. Compared to the rule-based TextBlob and lexicon-based VADER, BERT, a context-aware deep learning model, probably catches subtle sentiments and contextual meanings better. The differences show that whereas TextBlob and VADER may overstate neutrality or positivity, BERT is more sensitive to negative mood (Fig. 9).

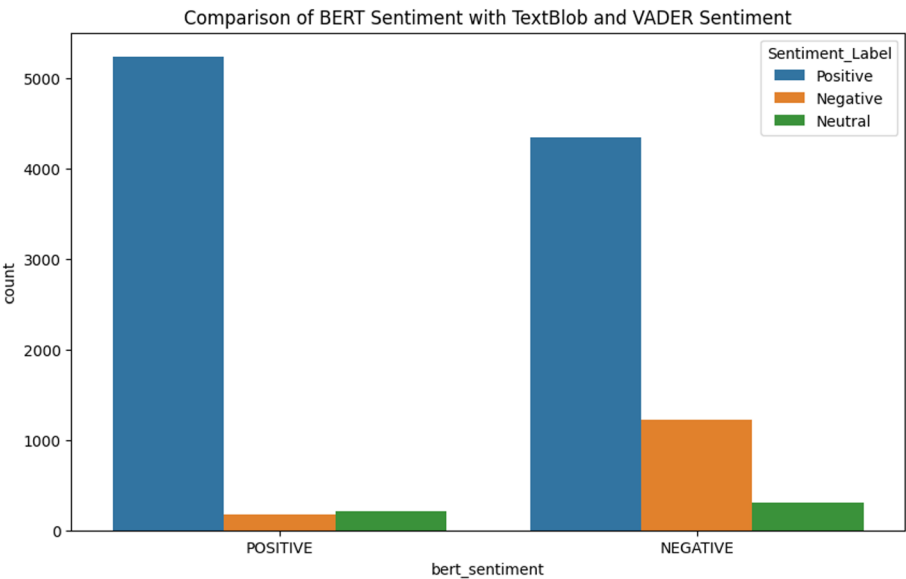
The BERT sentiment analysis categorization report on news stories about poverty and artificial intelligence shows a general accuracy of 56%, which is considered to be a mid-dling performance. The model accurately detects the majority of positive examples but misses some; it shows good accuracy for positive sentiment (0.93) but low recall (0.55). On the other hand, negative sentiment has a high recall (0.88) but a poor accuracy (0.21), meaning that although the model captures the majority of negative occurrences, it also incorrectly labels a large number of non-negative ones as negative. With an F1-score of 0.00, the model’s neutral sentiment identification is subpar, suggesting that it has trouble accurately identifying neutral items. With a weighted average F1-score of 0.62, the model is most effective when applied to positive emotion; negative and neutral classifications require work (Table 6).

Throughout three epochs, 9188 training samples and 2298 test samples were used to train the BERT sentiment analysis. Since the training loss dropped from 0.234 (Epoch 1) to 0.074 (Epoch 3), the model was able to learn from the data with effectiveness. Nevertheless, the validation loss at first stayed near the training loss (0.237 at Epoch 1), but by Epoch 3, it rose to 0.330, suggesting overfitting, in which the model does well on training data but has trouble generalizing to new data. Although the model’s ability to predict sentiment on test data is reasonable, regularization strategies, dropout modifications, or

**Table 5** Descriptive statistics of topics, clusters, BERT scores, and sentiment scores in AI-Poverty discussions

| Statistic | Topic  | Cluster | BERT_Topic | BERT_Score | Sentiment_Score |
|-----------|--------|---------|------------|------------|-----------------|
| Count     | 11,486 | 11,486  | 11,486     | 11,486     | 11,486          |
| Mean      | 1.693  | 2.193   | 1.174      | 0.943      | 0.712           |
| Min       | 0.000  | 0.000   | 0.000      | 0.500      | -1.000          |
| 25%       | 1.000  | 1.000   | 0.000      | 0.953      | 1.000           |
| 50%       | 2.000  | 2.000   | 1.000      | 0.991      | 1.000           |
| 75%       | 2.000  | 3.000   | 2.000      | 0.998      | 1.000           |
| Max       | 4.000  | 4.000   | 3.000      | 0.999      | 1.000           |
| Std       | 1.137  | 1.266   | 1.011      | 0.107      | 0.669           |

The table summarizes the distribution of topics, clusters, BERT classifications, and sentiment scores, showing that AI and poverty discussions are predominantly positive, with some polarization in policy and labor-related topics.



**Fig. 9** Comparison of BERT Sentiment with TextBlob and VADER Sentiment Note: The bar chart demonstrates the comparison of BERT sentiment classification with TextBlob and VADER sentiment labels, showing that BERT captures more nuanced sentiment, particularly detecting more negative articles, whereas TextBlob and VADER tend to classify more content as neutral or positive.

**Table 6** BERT sentiment classification report

| Category     | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Negative     | 0.21      | 0.88   | 0.34     | 1397    |
| Neutral      | 0.0       | 0.0    | 0.0      | 512     |
| Positive     | 0.93      | 0.55   | 0.69     | 9577    |
| Accuracy     |           |        | 0.56     | 11,486  |
| Macro avg    | 0.38      | 0.47   | 0.34     | 11,486  |
| Weighted avg | 0.8       | 0.56   | 0.62     | 11,486  |

The BERT sentiment classification report shows strong performance in detecting positive sentiment but challenges distinguishing negative and neutral sentiments, with a generic accuracy of 56%, emphasizing the need for further fine-tuning or additional contextual training

**Table 7** Training and validation loss

| Epoch | Training Loss | Validation Loss |
|-------|---------------|-----------------|
| 1     | 0.234000      | 0.237440        |
| 2     | 0.145700      | 0.243078        |
| 3     | 0.074400      | 0.330663        |

The table presents the training and validation loss across three epochs, showing progressive learning with decreasing training loss but a rising validation loss, indicating overfitting, which may impact the model’s ability to generalize on unseen sentiment classification tasks

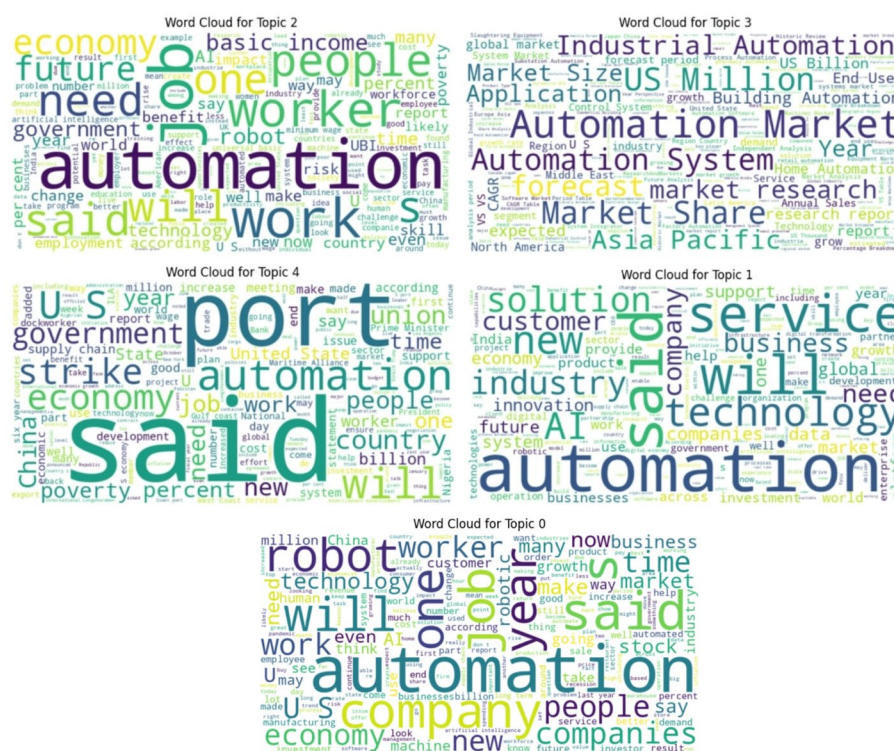
more training data might enhance its performance, as indicated by the final assessment loss of 0.3306 (Table 7).

Key topics in AI and poverty-related news are highlighted by the word clouds derived from LDA topic modeling, which highlight the workforce, business, policy, and economic aspects of automation. Topic 0 focuses on how automation and robots are being incorporated into sectors, influencing changes in labor and the economy. Topic 1 emphasizes automation’s role in market expansion and digital transformation by concentrating on AI-driven industrial innovation and company success. The economic and social ramifications of AI are covered in Topic 2, with special attention to how it will

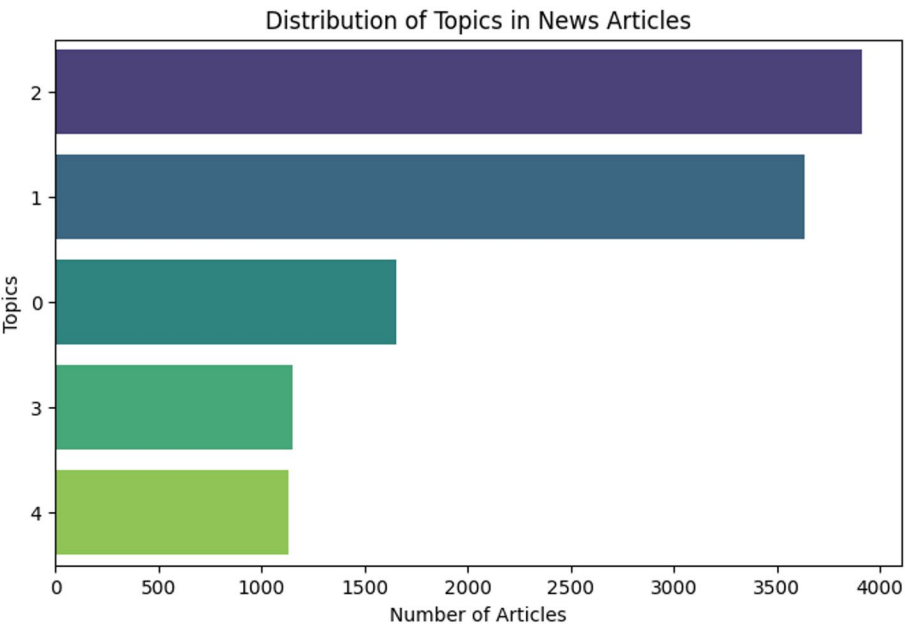
affect employment, governmental regulations, and debates over universal basic income (UBI). With financial estimates, industry research, and economic projections, Topic 3 focuses on regional AI adoption and worldwide market trends. Lastly, Topic 4 replicates the social and political opposition to automation by focusing on labor issues, union initiatives, and government interventions relating to AI. These insights reveal that AI and automation are widely debated, with a dominant narrative around economic growth yet significant concerns over labor market disruptions and policy responses (Fig. 10).

The distribution of subjects in news stories on poverty and artificial intelligence is depicted in the bar chart, which also shows which issues dominate media conversations. The highest representation of Topics 2 and 1 suggests that the media most commonly covers topics related to automation's effects on the economy and workforce, AI-driven company expansion, and technical breakthroughs. While Topics 3 and 4, which deal with market developments, industrial automation, and labor legislation, are covered less frequently, Topic 0, which is about robots, automation, and workforce transformation, is covered in a moderate number of articles. According to this distribution, industry-specific automation trends and government policy interventions receive somewhat less emphasis in media narratives than the impact of AI on jobs and economic structures (Fig. 11).

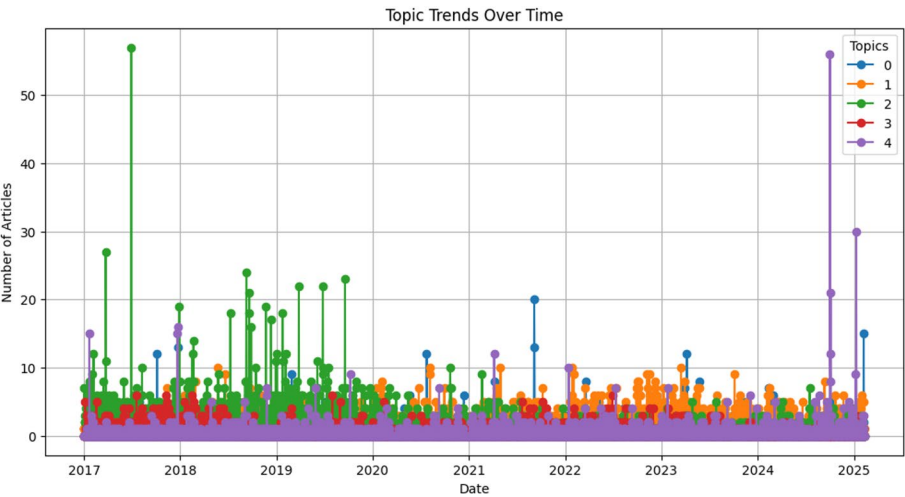
The scatter plot illustrates the changing nature of several AI and poverty-related themes between 2017 and 2025, revealing uneven media coverage. The surges in particular years suggest that media attention to AI-related concerns may have been impacted by particular events, technical developments, or regulatory changes. Significant peaks can be seen in Topics 2 (green) and 4 (purple), suggesting that conversations on the effects



**Fig. 10** Word Cloud for Topic 0, 1, 2, 3, and 4 Note: The word clouds represent key themes from AI and poverty-related news articles, emphasizing discussions on automation's economic impact, industry growth, workforce challenges, and government policies.



**Fig. 11** Distribution of Topics in News Articles Note: The bar chart displays the distribution of LDA-generated topics in news articles, emphasizing that economic and workforce-related themes conquer AI and poverty discourse, while market trends and policy discussions receive comparatively lower coverage.



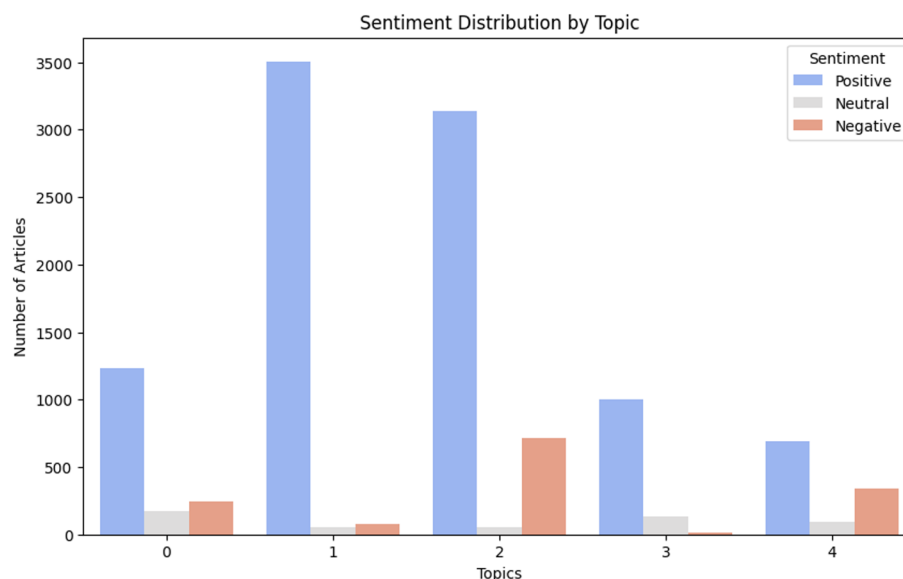
**Fig. 12** Topic Trends Over Time Note: The scatter plot demonstrates the evolution of topic coverage over time, revealing spikes in discussions on AI's economic and policy impacts, while business and workforce transformation topics remain consistently covered throughout the years.

of automation on the economy and labor force, as well as issues about policy, were more popular at particular times. Replicating the continuous conversation on AI-driven corporate expansion and industry changes, Topics 1 (orange) and 0 (blue) continue to receive comparatively consistent attention. The rise in Topic 4 (purple) following 2024 suggests that media attention is shifting toward economic policy, labor unions, and government actions in reaction to automation. These trends emphasize the dynamic nature of AI discussions, with varying emphasis on economic, workforce, and policy-related aspects over time (Fig. 12).

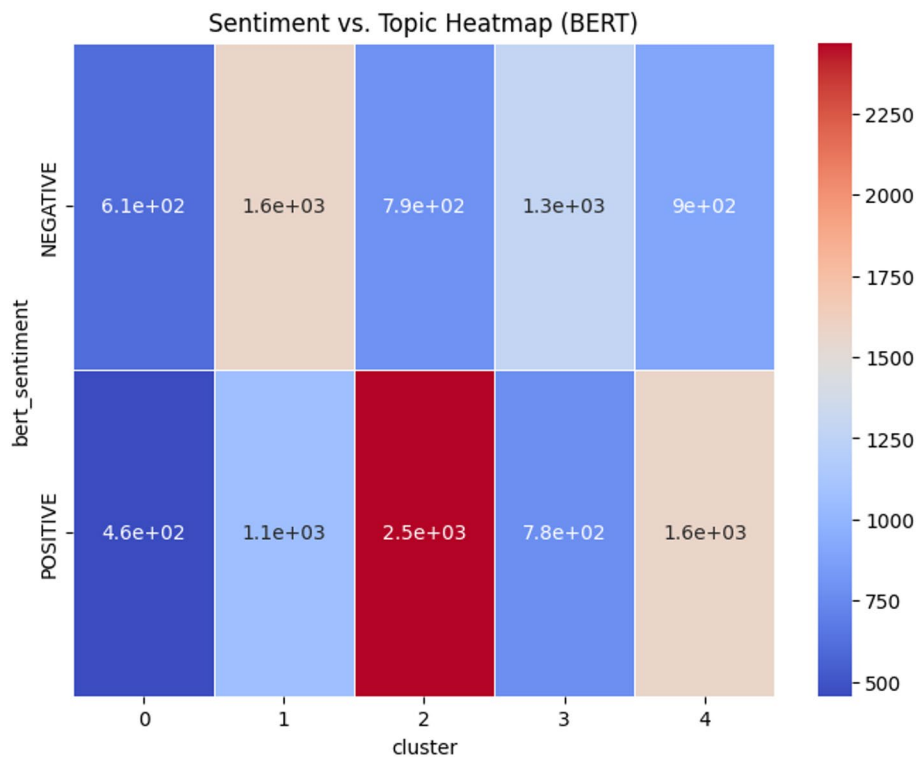
The bar graph highlights how various facets of automation and AI are seen in media coverage by displaying the sentiment distribution across various AI and poverty-related themes. The greatest number of articles with favorable sentiment (blue bars) are found on Topics 1 and 2, suggesting that most people have a good opinion on debates about AI-driven economic change and company growth. Concerns about job displacement and economic inequality are evident in Topic 2, which focuses on the workforce consequences of automation. This topic also has a large negative sentiment proportion (red bar). Compared to other issues, Topic 4 exhibits a comparatively greater level of negative emotion, suggesting media pessimism over labor union reactions and legislative changes connected to AI. Conversely, there is little neutral emotion (gray bars) on any of the subjects, indicating that AI conversation is politicized and frequently presented as either promising technological advancements or potential economic threats (Fig. 13).

A mostly pro-technology media narrative that emphasizes AI-driven job creation and economic growth is seen from the emotion vs. Topic Heatmap (BERT), which shows that automation's workforce effect and economic transformation (Topic 2) earn the highest favorable emotion. On the other hand, a divided attitude is evident in government policies and labor-related conversations (Topic 4), suggesting that labor union reactions and AI laws are hotly contested issues. Both excitement and reservations over the deployment of AI in sectors are reflected in the mixed views around AI-driven company development (Topic 1). While robotics and automation adoption (Topic 0) have lower generic sentiment intensity, suggesting that these topics are less contentious than policy and workforce debates, market trends and industrial automation (Topic 3) exhibit balanced sentiment, indicating more impartial, fact-based discussions (Fig. 14).

The Mean Absolute Error (MAE: 0.3170) and Root Mean Squared Error (RMSE: 0.4343), which show an average variance in sentiment ratings, indicate that the LSTM neural network utilized for sentiment prediction has moderate accuracy but limited predictive potential. The model barely explains 1.24% of the variance in real sentiment,



**Fig. 13** Sentiment Distribution by Topic Note: The bar chart shows the distribution of sentiment across AI and Poverty-related topics, revealing that business and economic transformation topics are mostly positive, while workforce and policy-related discussions contain higher negative sentiment, replicating concerns about automation's impact on labor markets and regulation.



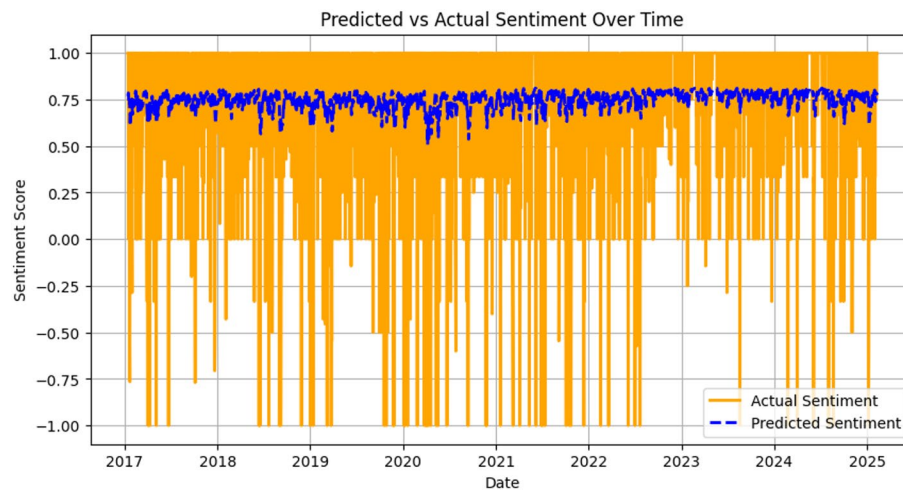
**Fig. 14** Sentiment v/s Topic Heatmap (BERT) Note: The Sentiment vs. Topic Heatmap (BERT) demonstrates the distribution of sentiment across AI-related topics, showing that workforce automation and economic transformation receive the supreme positive sentiment, while government policies and labor-related discussions display a mix of both positive and negative sentiment, indicating media polarization in AI policy debates.

**Table 8** Performance metrics

| Metric                         | Value  |
|--------------------------------|--------|
| Mean Absolute Error (MAE)      | 0.317  |
| Mean Squared Error (MSE)       | 0.1886 |
| Root Mean Squared Error (RMSE) | 0.4343 |
| R <sup>2</sup> Score           | 0.0124 |
| Pearson Correlation            | 0.1223 |

The table presents the LSTM model’s performance metrics, showing moderate accuracy but low predictive power, as signified by the low R<sup>2</sup> score (0.0124) and weak Pearson correlation (0.1223)

according to the low R2 value (0.0124), suggesting that it has trouble capturing intricate sentiment patterns over time. Additionally, the model has to be further optimized, whether by feature engineering, more training data, or hyperparameter tweaking, to boost prediction accuracy, as indicated by the poor correlation between the predicted and real sentiment scores, as indicated by the Pearson correlation (0.1223). One significant shortcoming of the current LSTM model is the removal of explanatory factors with high predictive value. While our method is based on sentiment time series, real-world drivers of poverty-related discourse are likely impacted by socioeconomic and policy variables. Incorporating such characteristics throughout subsequent iterations may result in stronger forecasting performance (Table 8).



**Fig. 15** Predicted v/s Actual Sentiment Over Time using LSTM Note: The LSTM sentiment prediction model shows moderate accuracy but weak predictive performance, with a low  $R^2$  score (0.0124) and weak Pearson correlation (0.1223), indicating that the model struggles to capture sentiment trends over time and may require further optimization through hyperparameter tuning or additional training data (Fig. 15).

## 5 Discussion of results

**RQ1** How do media narratives frame AI's role in poverty, and what are the dominant sentiment trends over time?

Sentiment analysis based on the application of BERT, VADER, and TextBlob reflects a consistent positive skew in media coverage of AI and poverty. This tendency is well supported by high average compound sentiment from VADER (0.6845) and a mean sentiment value of 0.1049, reflecting that AI is portrayed as an agent of transformation in economic development, job creation, and efficiency driven by automation. This is in line with Vinuesa et al. [77] research, which pointed out AI's role in advancing other Sustainable Development Goals (SDGs), particularly with its impact on economic systems.

However, the sentiment landscape is not uniformly positive. Coverage of policy discussion in the media tends to provoke negative sentiment against the backdrop of job loss, threat from automation, and growing socioeconomic inequality. Such a pattern converges with the fear reported by Lazer et al. [42], who opined that media framing of technological disruption can accumulate to societal fear and uncertainty.

Temporal analysis of topic trends indicates increased focus on the impact of the labor market and economic alterations induced by AI. Sentiment intensity differs highly between influential events, such as legislative initiatives and automation achievements. These event-related changes in sentiment are consistent with continuing work by Saputra et al. [68], which established that civic reaction to AI policy is prone to change and highly reactive to contemporary events. Polarization is a strong, definable element in labor-focused media communication, where opinions are influenced by opposing stakeholder positions, e.g., labor experts, politicians, and company managers.

The LSTM sentiment prediction model showed low predictive accuracy (MAE: 0.3170; RMSE: 0.4343;  $R^2 = 0.0124$ ), underscoring the complexity of anticipating sentiment in context-specific, dynamic environments. Such weaknesses are ably supported by studies carried out by Saif et al. [67], who highlighted that traditional models of sentiment are not accurate in anticipating nuances in socio-political and economic discourse, where temporality and context dominate.

In addition to typical LSTM designs, current research has suggested improvements for sentiment predictions. These include attention-based methods, hybrid models, and multimodal inputs that combine textual and exterior cues. For example, Singh et al. [73] used deep learning approaches combining LSTM and RNN to enhance sentiment categorization [20]. Similarly, Krugmann and Hartmann [39] showed that Large Language models like Llama 2 offer an unimaginable classification explanation.

The positive sentiment bias in the dataset may have an impact on both the sentiment classification findings and the LSTM model's forecasting accuracy. Sentiment forecasting models frequently assume reasonably balanced class distributions; in this case, the imbalance may have lowered sensitivity to negative or critical trends associated with poverty and social vulnerability. While this may indicate a real-world media bias, future research might look at resampling approaches, sentiment normalization, or class-weighted loss functions to overcome such imbalances and provide more robust modeling. While BERT and LSTM models displayed useful classification and prediction skills, there was some underperformance reported. This could be attributed to several factors: (1) inherent noise and contextual ambiguity in news data, which may reduce semantic clarity; (2) class imbalance in sentiment labels, which skews model sensitivity; and (3) temporal variation in language use and sentiment tone, which standard LSTM may not fully capture without more sophisticated temporal embedding. Future versions of the model may investigate sophisticated pretraining on domain-specific corpora or the use of attention processes to increase interpretability and accuracy.

One of the limitations observed in BERT was that it could not classify reliably in neutral sentiment, as evidenced by its low F1-score in the category. This has profound consequences when examining media narratives, where neutrality may signify balanced reporting, ambivalence, or editors' restraint. Not being able to reliably detect neutral sentiment can lead to an overestimation of perceived polarization or positivity, distorting our perception of political discourse. Because neutral sentiment often includes nuanced or mixed opinions, most particularly in highly complex issues like AI and poverty, this shortfall reduces the interpretive precision of the analysis. As Mohammad and Turney [53] previously noted, misclassification within the neutral class can skew sentiment aggregation, most particularly within corpora where neutrality constitutes a significant percentage of the tone. Subsequent research needs to account for calibration techniques or ensemble methods to counteract this bias.

**RQ2** What are the key themes in AI and poverty-related media discussions, and how do different sentiment analysis models interpret them?

Latent Dirichlet Allocation (LDA) topic modeling uncovered five dominant themes of AI and poverty media discussion: government policy, market drift, workforce transformation, enterprise expansion by AI, and robotics integration. These identify congruence with Binns [7], who reported that public discussion of AI gravitates towards labor disruption and regulation. Among the themes uncovered, economic development and workforce transformation were widely reported with a positive tone, while policy-related and labor union issues came across as more polarized.

Sentiment analysis across models revealed differing interpretations of the same themes. VADER detected a strong average sentiment score (mean: 0.6845), yet flagged more negative tones in discussions involving policy and labor issues. In contrast, BERT

reported highly positive sentiment (mean: 0.9433), suggesting an overall optimistic portrayal of AI's potential, particularly in sectors like automation and entrepreneurship. TextBlob gave more balanced scores, in line with its more moderate management of sentiment change. The contrast between tools is an indication of model-specific sensitivities and aligns with findings by Mohammad and Turney [53], which demonstrated that sentiment classifiers are prone to diverge when they classify affectively nuanced or contextually dense texts.

These model variations also reflect broader tensions in AI discourse. While BERT can exacerbate optimistic hopes due to its deep contextual embeddings, VADER's rule-based inference identifies more pointed sentiment shifts, especially in highly contested narratives. TextBlob's mid-level outputs suggest a flatter media tone representation, probably minimizing sentiment extremes. These variations underscore the necessity of using multiple models to triangulate for sentiment, especially when debating complex, ethically charged matters like labor rights and injustice.

Ultimately, since media news primarily depict AI in a positive, useful light, particularly with regards to innovation and economic productivity, there exist ongoing contentious issues of job loss, inequality, and ethics-driven regulation running deep below. This substantiates Fast and Horvitz's [15] observation that since media narratives often praise the potential of AI, they are fueled by context-bound pessimism and concern about its broader social implications.

## 6 Conclusion

This study uses sentiment analysis, topic modeling, and predictive modeling to give a thorough overview of AI's influence in media talks about poverty. According to sentiment analysis (VADER, TextBlob, and BERT), AI is primarily depicted in a positive light. This is especially true in narratives about worker automation and economic development. More divisiveness is seen in talks on policy and labor market issues, which emphasize both the advantages and disadvantages of adopting AI. The findings of the topic modeling indicate that five main topics dominate media discourse, with a notable emphasis on developments in industrial automation, workforce change, and AI-driven corporate expansion. Despite the widespread portrayal of AI as a vehicle for economic advancement, worries about job displacement, regulation, and digital inequality continue to exist, which causes varying attitude patterns across various news sources. All things considered, this study shows how AI's view in the discourse around poverty is changing, highlighting the necessity of fair media representations and legislative measures that take into account both the advantages and the socioeconomic consequences of automation. Future studies might examine regional differences in attitudes toward AI, the efficacy of policies, and the actual economic effects of AI-driven automation to get a better understanding of its potential to reduce poverty and promote sustainable development. The policy implications of this study are for policymakers, media analysts, and scholars. They offer insights into AI news sentiment as an indicator of public awareness of poverty and sustainability that can guide policy, media framing analysis, and scholarly research on technology and sustainable development. In conclusion, this study contributes freshly by linking sentiment trends in AI news to SDG 1 (No Poverty). Unlike existing studies that either focus on sentiment analysis or on sustainability reporting alone, this study links the two areas to indicate the correlation between public opinion and

international development efforts. By unveiling sentiment trends around AI in poverty reduction contexts, this approach offers a new way to quantify communicative alignment between tech development and sustainability goals.

However, this study is not without limitations. First, even though GDELT provides global coverage, it may be media presentation and language use-biased. Second, the sentiment models used-especially lexicon-based models, cannot hold nuances of contextual or cultural interpretations of poverty and AI. Third, the limited predictive ability of the LSTM model suggests that future studies can explore hybrid or transformer-based prediction models. Future research can also benefit from employing multilingual data and cross-platform comparisons for improving representativeness and robustness. Although LSTM models are technically sophisticated, our investigation found modest predictive ability ( $R^2 = 0.0124$ , Pearson correlation = 0.1223). The primary cause is the univariate character of the input data sentiment ratings obtained exclusively from media headlines, which may fail to represent the complex processes shaping poverty discourse. The absence of external factors, such as economic indicators (e.g., GDP growth, inflation, unemployment) or policy announcements (e.g., social welfare changes), may limit the model's capacity to detect important temporal trends. Future research might benefit from multimodal LSTM designs that use such factors to increase contextual awareness and prediction accuracy. And also, the data for 2025 only covers headlines up to February 8, therefore, it only provides a partial picture of the year's conversation. As a result, interpretations of 2025 sentiment patterns should be regarded as tentative and not indicative of the whole calendar year.

This study contributes to the growing body of interdisciplinary research at the intersection of artificial intelligence, media discourse analysis, and sustainable development. By coupling sentiment analysis (via BERT, VADER, and TextBlob), topic modeling (LDA), and time-series forecasting (LSTM), we offer a new empirical framework to evaluate how AI is constructed in global media reports on poverty. The findings indicate a wide pro-bias on AI sentiment with moderation of polarization of labor and policy issues. Notably, this paper fills an important schism by directly connecting sentiment patterns to framing SDG1 (No Poverty), presenting how public discourse reiterates or resists development narratives. In doing so, we provide a reproducible, scalable approach to tracking the public's view of emerging technology in the context of sustainability discourse, a space of growing importance to policymakers, media analysts, and AI ethicists.

#### **Author contributions**

R.K.M.R. conceptualized the study, conducted the literature review, and performed data analysis. M.V. supervised the research, contributed to the methodology, and provided critical revisions. S.S. contributed to data collection, manuscript writing, and result interpretation. All authors reviewed and approved the final manuscript.

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#### **Availability of data and materials**

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

#### **Declarations**

##### **Ethics approval and consent to participate**

Not Applicable.

##### **Consent for publication**

The authors have agreed to submit and have approved the manuscript for submission.

### Competing interests

The authors declare no competing interests.

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